Pandelis Zembashis

CTO | Staff-Level Engineer | Applied AI, Agentic Coding & Platform Reliability

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EXECUTIVE SUMMARY

CTO and staff-level engineer building AI-native products from 0 to 1 through scale across developer tools, enterprise products, fintech, mobility, and IoT. At Cosine, lead product and engineering for a sovereign AI coding platform selected for the UK Government's £500m Sovereign AI programme, with AIRR supercompute backing for British-owned model development. Hands-on across product experience, tool orchestration, execution infrastructure, observability, and air-gapped private deployment. Previously scaled 11x.ai from 0 to $1M ARR in under six months and have shipped across React, TypeScript, Node.js, GraphQL, Go, WebAssembly, Python, Kubernetes, and AWS.

PROFESSIONAL EXPERIENCE

Cosine | CTO | Oct. 2023 – Present

Sovereign agentic coding product spanning developer tooling, enterprise deployment, and emerging desktop and IDE surfaces.

- Own end-to-end product and engineering for Cosine's sovereign AI coding platform, selected for the UK Government's £500m Sovereign AI programme and awarded 500,000 AIRR GPU hours on Isambard-AI.
- Architected Cosine CLI as a single Go binary with browser use, internet research, memories, and bundled ripgrep, enabling portable developer tooling and lower-friction enterprise deployment.
- Designed the event-bus agent loop, custom Harness, and cloud agents running on scalable Kubernetes and gVisor-based workers to support reliable long-running execution, tool use, and state propagation.
- Built foundations for desktop and VS Code experiences while keeping shared runtime contracts explicit across CLI, UI, worker, and extension-oriented surfaces.
- Shaped VPC, on-prem, and air-gapped deployment patterns for regulated and sensitive environments where code, model calls, and operational data cannot leave customer infrastructure.

11x.ai | Head of Engineering | Apr. 2023 – Sep. 2023

Fastest-growing UK AI startup from 0 to $1M ARR in under six months. TechCrunch coverage.

- Scaled the codebase and engineering organization behind Alice as 11x.ai grew from 0 to $1M ARR in under six months.
- Scaled outbound email and event-driven automation systems that processed millions of email chains and billions of tokens.
- Led product-minded engineering trade-offs across AI worker workflows, customer feedback, operational bottlenecks, and rapid iteration during seed-to-Series A growth.
- Established onboarding and operating rhythms that kept the engineering team shipping through high-growth ambiguity.

Papaya | Technical Lead | Feb. 2022 – Mar. 2023

Marketplace and fleet platform helping electrify commercial and last-mile logistics. TechCrunch coverage.

- Led delivery of a fleet management platform in Next.js, React Query, and Tailwind CSS for light electric vehicle operations.
- Modernized the product API layer by introducing GraphQL into the Ruby on Rails backend, improving iteration across product and engineering.
- Established development cadences, product intake, and stakeholder feedback loops that improved execution across engineering and product.

Weezy (acquired by Getir) | Senior Software Engineer | Aug. 2021 – Jan. 2022

Rapid grocery delivery startup building warehouse, ordering, and customer systems.

- Took over the core ordering and processing monolith in Ruby on Rails, owning day-to-day delivery and forward planning during a high-growth phase.
- Built warehouse operations systems and supporting AWS and Heroku infrastructure with a focus on scale, availability, incident response, and security.

finleap connect | Senior Software Engineer | Nov. 2020 – Aug. 2021
White-label fintech platform powering digital banking products and apps.

- Built GraphQL APIs on NestJS and cross-platform clients in React Native, Apollo, Node.js, and TypeScript for a multi-tenant fintech platform.
- Led alerting and observability improvements while supporting migration from ECS to Amazon Lambda.
- Owned client-specific delivery across the stack, from GraphQL resolvers and mutations to customer-facing mobile screens.

IoT CONNCTD GmbH | Full Stack Engineer | Jul. 2019 – Nov. 2020
Cloud IoT platform unifying device protocols behind a GraphQL API.

- Built frontend interfaces, SDKs, and internal tooling for a cloud IoT platform exposing a unified GraphQL API across multiple device protocols.
- Designed authentication, update, and upstream synchronization flows using TypeScript, React, MobX, and Apollo GraphQL.

Substrakt | Software Developer | Aug. 2018 – Jun. 2019
Digital product agency building bespoke publishing, arts, and cultural web platforms.

- Built bespoke WordPress features, search experiences, and internal tooling for design-led client websites.
- Modernized the JavaScript toolchain from Gulp to Webpack, improving development and deployment workflows.

D4 Software | Software Engineer | Jun. 2017 – Jul. 2018
Product engineering consultancy building client platforms, tools, and cloud infrastructure.

- Built client-facing features and internal tools across product and platform engagements using JavaScript and .NET.
- Maintained AWS infrastructure, serverless services, databases, and virtual machines while modernizing the frontend stack with Webpack, TypeScript, and Vue.js.

SELECTED WRITING, TALKS & PUBLIC WORK

Developer Tooling: Cosine CLI 1.16 Update: public write-up on single-binary distribution, internal search, and developer-tool architecture.
Applied AI Writing: Multi-Agents; Enterprise Demands; Parallelising Development
Talks & Media: AWS Startups - The Future Of Work; Directus TV - Edge Computing with Pandelis; Git Intro Workshop
Public Portfolio: pa.ndel.is; Devpost; Product Hunt launches

EDUCATION

Birmingham City University | Bachelor's Degree, Computer Science with Year in Industry | 2015 – 2019

AREAS OF EXPERTISE

Applied AI Product Engineering: LLM-powered applications, agentic coding workflows, full-stack product architecture, developer experience
Agent & Tool Orchestration: CLI tooling, browser use, memory systems, event-driven architectures, long-running execution, IDE integrations
Production Platform & Reliability: Go, Python, AWS, Kubernetes, gVisor workers, observability, logging, tracing, incident-oriented debugging
Sovereign Deployment: React, TypeScript, Node.js, GraphQL, VPC, on-prem, air-gapped, customer-facing delivery